

1. Which members of the Circle class are encapsulated?

The radius and PI are encapsulated since they are private and only visible to the class.

2. What is the requirement for the name of a constructor for a class?

The constructor needs to be named exactly like the class.

3. Explain the difference between the private and public access specifiers.

Private members are limited to accessing only within the class. On the other hand, Public members are accessible from other classes.

4. Is the statement above valid or invalid? Provide valid justification.

It is valid only in the case where radius is public. It is invalid when radius is private.

5a. What is the name of the class?

Roo

5b. What is the name of the data member?

x

5c. Name the accessor method

getX

5d. Name the modifier method

setX

5e. Name the supporting method

factor

5f. What is the name of the constructor?

Roo

5g. What is the total number of method members?

4

6. Explain the difference between a class and an object.

A class is a design and an object is the realization of that design.

9a. Which data member is a constant?

z

9b. Which data members are variables?

x and y

9c. Which data members are instance members?

y

9d. Which data members are class members?

x and z

11. Describe the differences between methods that are overridden and those that are overloaded

Overloading is using the same method name with different parameters in the same class, while Overriding is providing a new definition for a class's parent method in the child class.